

	1	10	20	30	40	50	60
Escherichia_coli_groL	MAAKDVKFGND	ARVKMLR	GVNVLAD	AVKVTLGPKGRNVVLDKS	FGAPTITK	DGVS	VAREI
Thermus_thermophilus_groL	.MAKILVFDEA	ARRALER	GVNAVANA	VKVTLGPRGRNVVLEKK	FGSPTITK	DGVT	VAKEV
Bifidobacterium_animalis_groL	.MAKITEYDEE	ARQGMLA	GLDKLD	TVKVTLGPKGRNVVLDKT	YGAPTITN	DGVS	IAKEI
Corynebacterium_glutamicum_groL	.MAKLIAFDQD	AREGILR	GVDALANA	VKVTLGPRGRNVVLDKA	FGGFLVTN	DGVT	IARDI

	70	80	90	100	110	120
Escherichia_coli_groL	ELEDKFEENM	GAQMVEVAS	KAND	AGDGTTTATVLAQAIIT	EGLKAVA	AGMNPMDLKRGI
Thermus_thermophilus_groL	ELEDHLENIG	AQLLKEVAS	KTNDV	AGDGTTTATVLAQAIIV	REGLKNVA	AGANPLALKRGI
Bifidobacterium_animalis_groL	DLEDPEERIG	AEIVKEVAK	RTDDV	AGDGTTTATVLAQSLV	HEGLKNVV	AGSNPIALRRGI
Corynebacterium_glutamicum_groL	DLEDPEENLGA	QLVKSVA	KTNDI	AGDGTTTATLLAQALIA	EGLRNV	AGANPEMELNKRGI

	130	140	150	160	170	180
Escherichia_coli_groL	DKAVTAAVEEL	KALSVPCSD	SRKIAQVGT	ISANSDET	VGKLI	IAEAMD
Thermus_thermophilus_groL	EKAVEAAVEKI	KALAIIPVED	RKAIEEVA	TISAN.D	PEVGKLI	ADAMEK
Bifidobacterium_animalis_groL	EKASDALVKQL	VASAKPVE	TKEQIAATA	TISAG.D	PEVGKLI	AEALDK
Corynebacterium_glutamicum_groL	SAAAEKTLLEL	KARATEV	SDTKEIANVA	TVSSR.D	EVVGEI	VAAAMEK

	190	200	210	220	230	240
Escherichia_coli_groL	TGLQDELDVV	EGMDFDRGYL	SPYFIN	KPTEGAVE	LESFFILL	ADKKISN
Thermus_thermophilus_groL	KSLETELKKFV	EGYQFDKGYI	SPYFVT	NPETMEAV	LEDAFILI	VEKKVSN
Bifidobacterium_animalis_groL	NRFGLDLDFTE	GMREDKGYI	SPYFVT	NAEDQTAV	LDDPYILL	TSSKVVSS
Corynebacterium_glutamicum_groL	QSIETALEVTE	GISFDDKGYL	SPYFIN	DNDTQQAV	LDNPAVLL	VRNKRIS

	250	260	270	280	290	300
Escherichia_coli_groL	AKAGKPLLI	IAEDVEGE	ALATLVVN	TMRGIVKVA	AVKAPFG	FGDRRK
Thermus_thermophilus_groL	AQTGKPLLI	IAEDVEGE	ALATLVVN	NKLRGTL	SVA	AVKAPFG
Bifidobacterium_animalis_groL	MKTGKPLLI	IAEDVDGE	ALPTLILNN	IRGTFK	SCAVKAP	FGDRRK
Corynebacterium_glutamicum_groL	VESNRPLLI	IAEDVEGE	PLQTLVVNS	IRKTIKVV	AVKSPY	FGDRRK

	310	320	330	340	350	360
Escherichia_coli_groL	ISEEIGMELE	KATLEDL	GQAKRVV	TNKDTT	ITIDGV	GEEAAT
Thermus_thermophilus_groL	ISEEIGFKLEN	ATLSMLGR	AERVRI	ITKDETT	IVGGKK	KKEDIE
Bifidobacterium_animalis_groL	VSEDLGLKL	DSIDLSV	FGTAKKVI	IVSKDETT	IVSGGK	SKEDVA
Corynebacterium_glutamicum_groL	VDPEVGINL	NEAGEEV	FGTAR	ITVS	KDETTI	VDGAGS

	370	380	390	400	410	420
Escherichia_coli_groL	DREKLQERV	AKLAGGV	AVIKVGA	ATEVEM	KEKKAR	VEDAL
Thermus_thermophilus_groL	AREKLQERL	AKLAGGV	AVIRVGA	ATETEL	KEKKHR	FEDAL
Bifidobacterium_animalis_groL	DREKLQERL	AKLAGGV	AVIKVGA	ATEVEA	KERKHR	IEDA
Corynebacterium_glutamicum_groL	DREKABERL	AKLAGG	IAVIRV	GAAATE	EVNDR	KLRVED

	430	440	450	460	470
Escherichia_coli_groL	RVASKLADLR	GQNE.D	QNVGI	KVALRAME	AFILRQ
Thermus_thermophilus_groL	RAISAVEELI	KKLEGG	DEATG	AKIVRR	ALEBPARQ
Bifidobacterium_animalis_groL	QAAEKVEK	DFN.LEG	DEATG	AAIVFSG	IEAFIKQIA
Corynebacterium_glutamicum_groL	QIAETLKAY	AEFE	EGDQKV	GVRALATA	LGRPAYW

	480	490	500	510	520	530
Escherichia_coli_groL	YNAATEEY	GNMIDM	GILDP	TKVTRS	SALQV	AA
Thermus_thermophilus_groL	FNAATGEF	VDMVEAG	IVDPA	KVTRS	SALQNA	AS
Bifidobacterium_animalis_groL	FNAATDTY	EDLMAG	VTD	PKVTRS	SALQNA	AS
Corynebacterium_glutamicum_groL	FNAATLEY	GNLIND	GVIDP	VKTHS	AVNAT	SV

	540
Escherichia_coli_groL	MGGMGGMGGMM
Thermus_thermophilus_groL	AGDMD....F
Bifidobacterium_animalis_groL	.QDMG....Y
Corynebacterium_glutamicum_groL	..HHH....H